

2.4.1 Splitting Methods

Let $H(\mathbf{q}, \mathbf{p}) = H_1(\mathbf{q}, \mathbf{p}) + H_2(\mathbf{q}, \mathbf{p})$ have flow map $\mathcal{F}_h$, and indicate by $\mathcal{F}_h^1, \mathcal{F}_h^2$ the flow maps for the systems with Hamiltonians H_1, H_2 , respectively. The proposal is that the map

$$\mathcal{G}_h = \mathcal{F}_h^1 \circ \mathcal{F}_h^2$$

is an approximation of $\mathcal{F}_h$. For this to be a first order method, we need at least

$$\|\mathcal{G}_h(\mathbf{u}) - \mathcal{F}_h(\mathbf{u})\| \leq C(\mathbf{u})h^2.$$

Expand $\mathcal{F}_h(\mathbf{u})$ in a Taylor series:

$$\mathcal{F}_h(\mathbf{u}) = \mathbf{u} + h\mathbf{J}\nabla H + \mathcal{O}(h^2) = \mathbf{u} + h(\mathbf{J}\nabla H_1 + \mathbf{J}\nabla H_2) + \mathcal{O}(h^2). \quad (2.24)$$

On the other hand,

$$\mathcal{F}_h^1(\mathbf{u}) = \mathbf{u} + h\mathbf{J}\nabla H_1(\mathbf{u}) + \mathcal{O}(h^2), \quad \mathcal{F}_h^2(\mathbf{u}) = \mathbf{u} + h\mathbf{J}\nabla H_2(\mathbf{u}) + \mathcal{O}(h^2).$$

Composing the maps we have

$$\mathcal{F}_h^1 \circ \mathcal{F}_h^2(\mathbf{u}) = \mathbf{u} + h\mathbf{J}\nabla H_2(\mathbf{u}) + h\mathbf{J}\nabla H_1(\mathbf{u} + h\mathbf{J}\nabla H_2(\mathbf{u})) + \mathcal{O}(h^2).$$

Assuming H_1 is C^2 , we have

$$\mathcal{F}_h^1 \circ \mathcal{F}_h^2(\mathbf{u}) = \mathbf{u} + h(\mathbf{J}\nabla H_2(\mathbf{u}) + \mathbf{J}\nabla H_1(\mathbf{u})) + \mathcal{O}(h^2),$$

which agrees with the expansion (2.24) of the flow map through the terms of first order. Thus the splitting method does indeed provide a second order local approximation of the flow map.

Example 2.4 Let

$$H_1(\mathbf{q}, \mathbf{p}) = \mathbf{p}^T \mathbf{M}^{-1} \mathbf{p} / 2, \quad H_2(\mathbf{q}, \mathbf{p}) = U(\mathbf{q}),$$

then a splitting method for $H = H_1 + H_2$ may be obtained by determining the flow maps for each of the two parts. For H_1 we have the differential equations

$$\dot{\mathbf{q}} = \mathbf{M}^{-1} \mathbf{p}, \quad \dot{\mathbf{p}} = \mathbf{0}.$$

The second equation tells us that $\mathbf{p}$ is constant, so it is fixed at its initial value, whereas the first equation says that $\mathbf{q}$ evolves on a linear path, hence its flow map is

$$\begin{aligned} \mathbf{Q} &= \mathbf{q} + h\mathbf{M}^{-1} \mathbf{p}, \\ \mathbf{P} &= \mathbf{p}. \end{aligned}$$